

# RENATO SPRENGER-VUKOVIĆ

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## SUMMARY

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PhD candidate in Computer Science researching large language models, information extraction, and knowledge structuring. Author of multiple peer-reviewed publications at top-tier venues and experienced in teaching and supervising undergraduate and graduate research. Furthermore, I have experience in applying my research to real-world problems through freelance projects in industry, demonstrating the practical impact of my work.

## EDUCATION

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**PhD in Computer Science**, Heinrich Heine University 2023 – 2027 (expected)

Large Language Models, Information Extraction, Task-oriented Dialogue

**Master of Computer Science**, Heinrich Heine University 2019 – 2022

**Bachelor of Computer Science**, Minor in Physics, Heinrich Heine University 2015 – 2019

## EXPERIENCE

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**Doctoral Researcher** Jan 2023 – Jan 2027

Heinrich Heine University Düsseldorf

*Düsseldorf, Germany*

- Conducted doctoral research on large language models and information extraction, improving ontology construction accuracy by 50% using text-to-SQL.
- Supervised and taught Bachelor's and Master's thesis projects and courses, respectively.

**Freelance Machine Learning Engineer** Feb 2024 – Nov 2024

Deutscher Versand Service

*Remote*

- Designed and deployed a GDPR-compliant LLM-based customer service system, reducing manual workload.

**Freelance Data Analyst** Jan 2025 – May 2025

Westdeutscher Rundfunk

*Remote*

- Developed a statistical filtering pipeline to detect anomalous streaming activity, reducing daily variance by up to 50%.

## SELECTED PUBLICATIONS

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**Text-to-SQL Task-oriented Dialogue Ontology Construction** — Vukovic et al. — TACL 2026

**Post-Training Large Language Models via Reinforcement Learning from Self-Feedback** — Van Niekerk et al. — Preprint arXiv 2025

**Dialogue Ontology Relation Extraction via Constrained Chain-of-Thought Decoding** — Vukovic et al. — SIGDIAL 2024

**Less is More: Local Intrinsic Dimensions of Contextual Language Models** — Ruppik et al. — NeurIPS 2025

**Dialogue Term Extraction using Transfer Learning and Topological Data Analysis** — Vukovic et al. — SIGDIAL 2022

## SKILLS

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|-----------------------------|---------------------------------------------------------------------------------------|
| <b>Programming</b>          | Python (advanced), Java, C++                                                          |
| <b>Machine Learning</b>     | PyTorch, HuggingFace, structured prediction, model evaluation, fine-tuning, prompting |
| <b>Mathematics</b>          | Probability, statistics, linear algebra, hypothesis testing, optimization             |
| <b>Software Engineering</b> | Object-oriented design, test-driven development, Git                                  |
| <b>Cloud</b>                | Google Cloud Platform, Microsoft Azure                                                |